MINISTERUL EDUCAȚIEI ȘI CERCETĂRII AL REPUBLICII MOLDOVA

Universitatea Tehnică a Moldovei

Facultatea Calculatoare, Informatică şi Microelectronică

Departamentul Inginerie Software și Automatică

RAPORT

Lucrare de laborator Nr. 7
Disciplina: ,,Tehnici de Programare”

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Introducere

Scopul lucrării: constă în însușirea și consolidarea tehnicilor de control al versiunilor utilizând Git pentru gestionarea proiectelor software, dezvoltarea abilităților de depanare pas cu pas a aplicațiilor Python prin citirea și interpretarea corectă a mesajelor de eroare, precum și utilizarea istoricului de commit-uri Git ca jurnal structurat pentru documentarea procesului de remediere a codului sursă.

Sarcina lucrării: presupune configurarea mediului de lucru prin conectarea depozitului local Git exercitiu-git la platforma GitHub, identificarea și remedierea sistematică a celor 10 erori din scriptul inventar.py, documentarea fiecărei reparații prin commit-uri individuale cu mesaje explicative, iar în final executarea uneltei pentru generarea fișierului raport_inventar.json și încărcarea istoricului complet pe GitHub prin comanda git push.

Mersul Lucrarii:

Partea 1: Lucrarea A. GitHub, pas cu pas

```
(.venv) PS D:\univer\Python projects> git --version
git version 2.56.0.windows.3
(.venv) PS D:\univer\Python projects> git config --global user.name "Zlatin Nichita"
(.venv) PS D:\univer\Python projects> git config --global user.email "zlatinnichita2006@gmail.com"
(.venv) PS D:\univer\Python projects> git config --global init.defaultBranch main
(.venv) PS D:\univer\Python projects> git config --global pull.rebase false
(.venv) PS D:\univer\Python projects> git config --global --list
user.name=Zlatin Nichita
user.email=zlatinnichita2006@gmail.com
init.defaultbranch=main
pull.rebase=false
```

Figura 1.1: **A2. Git pe calculator: verifici și te prezinți**

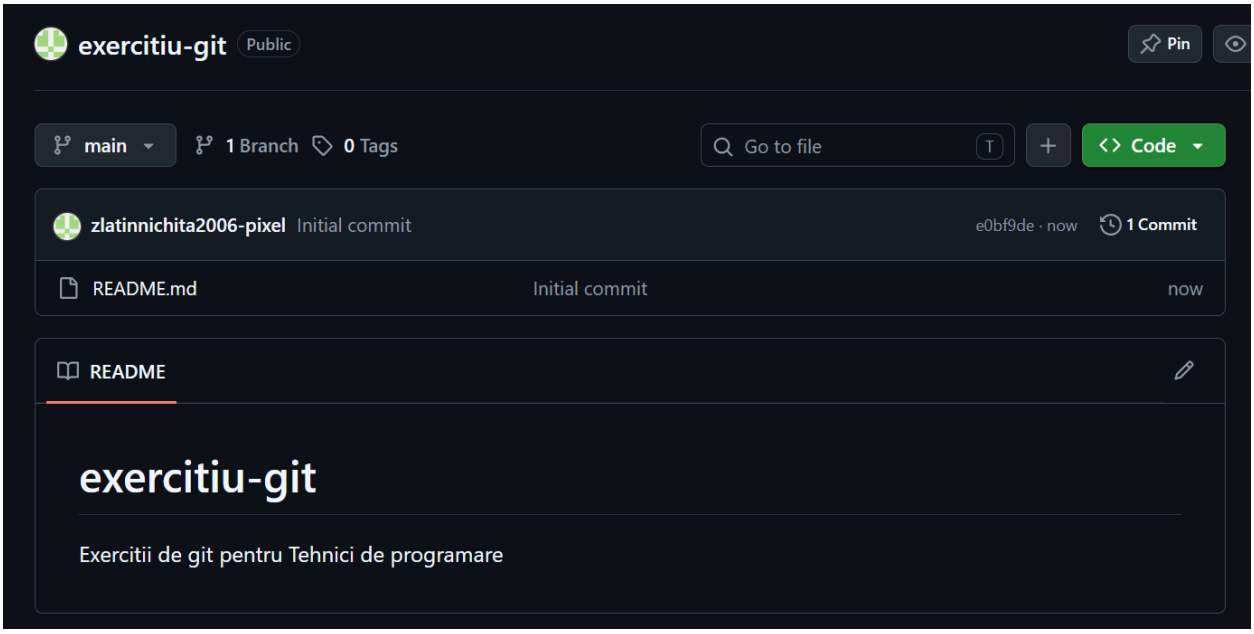


Figura 1.2: **A3. Primul depozit, creat pe GitHub**

```
(.venv) PS C:\Users\User> git clone https://github.com/zlatinnichita2006-pixel/exercitiu-git.git
Cloning into 'exercitiu-git'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

(.venv) PS C:\Users\User> cd exercitiu-git

(.venv) PS C:\Users\User\exercitiu-git> dir


    Directory: C:\Users\User\exercitiu-git


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        10/27/2026   2:12 PM             64 README.md


(.venv) PS C:\Users\User\exercitiu-git>
```

Figura 1.3: **A4. Îl aduci pe calculator: git clone**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git add salut.py

(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   .idea/.gitignore
        new file:   .idea/exercitiu-git.iml
        new file:   .idea/inspectionProfiles/profiles_settings.xml
        new file:   .idea/modules.xml
        new file:   .idea/vcs.xml
        new file:   salut.py

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .idea/misc.xml
```

Figura 1.4: **A5. Prima ta fotografie: status, add**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git commit -m "Adaug salut.py"
[main 2434ab4] Adaug salut.py
 6 files changed, 40 insertions(+)
 create mode 100644 .idea/.gitignore
 create mode 100644 .idea/exercitiu-git.iml
 create mode 100644 .idea/inspectionProfiles/profiles_settings.xml
 create mode 100644 .idea/modules.xml
 create mode 100644 .idea/vcs.xml
 create mode 100644 salut.py
```

Figura 1.5: **A5. Prima ta fotografie: commit**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git push
info: please complete authentication in your browser...
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 16 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 1.43 KiB | 731.00 KiB/s, done.
Total 10 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/zlatinnichita2006-pixel/exercitiu-git.git
   e0bf9de..2434ab4  main -> main

(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git>
```

Figura 1.6: **A6. O trimiți pe GitHub: git push**

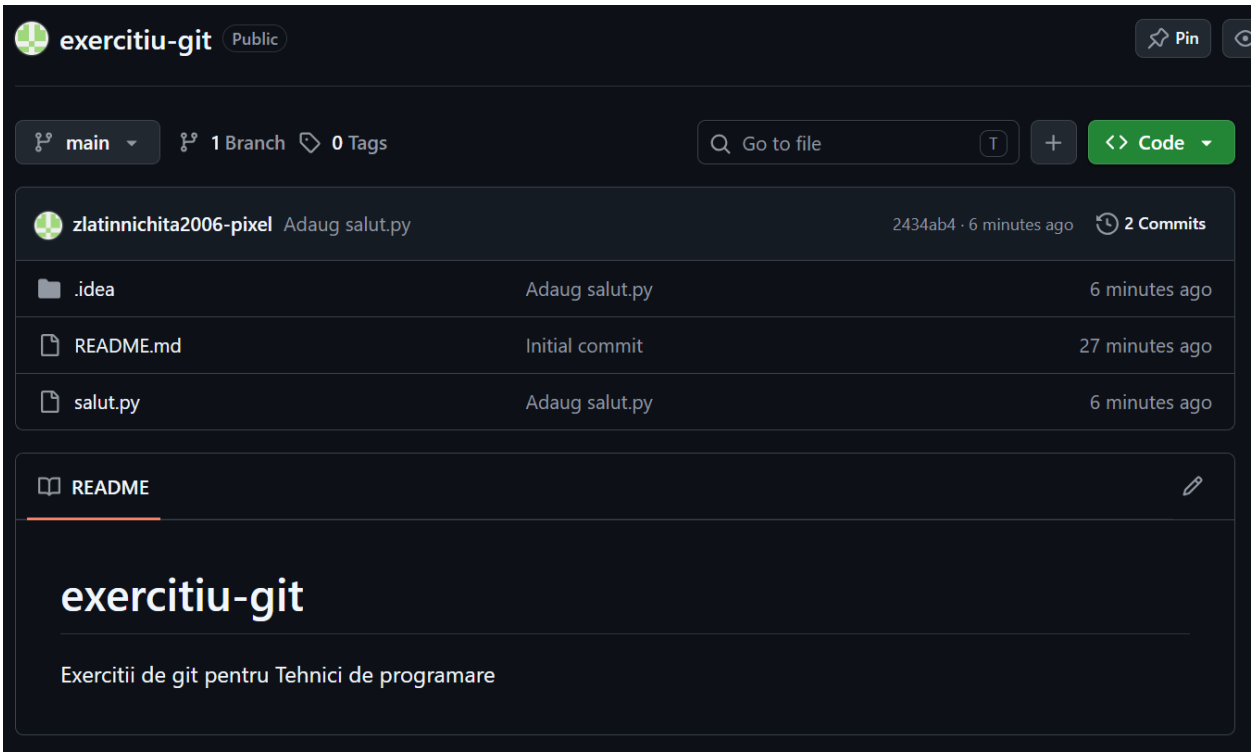


Figura 1.7: **A6 Github pe browser**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1015 bytes | 59.00 KiB/s, done.
From https://github.com/zlatinnichita2006-pixel/exercitiu-git
   2434ab4..bcd894a  main       -> origin/main
Updating 2434ab4..bcd894a
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> type README.md
# exercitiu-git
Exercitii de git pentru Tehnici de programare
Aici exersez git
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git>
```

Figura 1.8: **A7. O schimbare făcută pe GitHub, adusă înapoi: git pull**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> python -m venv .venv
Unable to copy 'C:\\Program Files\\WindowsApps\\PythonSoftwareFoundation.Python.3.13_3.13.3824.0_x64__qbz5n2kfra8p0\\Lib\\venv\\scripts\\nt\\venvlau
ncher.exe' to 'C:\\Users\\User\\PycharmProjects\\exercitiu-git\\.venv\\Scripts\\python.exe'
Error: Command '['C:\\Users\\User\\PycharmProjects\\exercitiu-git\\.venv\\Scripts\\python.exe', '-m', 'ensurepip', '--upgrade', '--default-pip']' re
turned non-zero exit status 1.
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git status --short
?? .idea/misc.xml
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git status --short
AM .gitignore
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git add .gitignore
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git commit -m "Ignor mediul virtual"
[main 53027d6] Ignor mediul virtual
 1 file changed, 4 insertions(+)
 create mode 100644 .gitignore
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 387 bytes | 387.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/zlatinnichita2006-pixel/exercitiu-git.git
   bcd894a..53027d6  main -> main
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git>
```

Figura 1.9: **A8. Ce NU urcă pe GitHub: .gitignore**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git log --oneline
53027d6 (HEAD -> main, origin/main, origin/HEAD) Ignor mediul virtual
bcd894a Update README.md
2434ab4 Adaug salut.py
e0bf9de Initial commit
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git diff
diff --git a/salut.py b/salut.py
index 2354a2f..9ca4b79 100644
--- a/salut.py
+++ b/salut.py
@@ -1 +1,2 @@
-print("salut, github!")
\ No newline at end of file
+print("salut, github!")
+print("al doilea")
\ No newline at end of file
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git restore salut.py
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git>
```

Figura 1.10: **A9. Citești istoria și schimbările: log și diff**

Lincul pentru exercitiul-git: ,, https://github.com/zlatinnichita2006-pixel/exercitiu-git “

Lincul pentru exercitiul-2: ,, https://github.com/zlatinnichita2006-pixel/exercitiu-2.git “

Partea 2: Lucrarea B. Unealta stricată, reparată cu git

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-2> cd C:\Users\User\PycharmProjects\exercitiu-git
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> git pull
Already up to date.
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> mkdir depanare


    Directory: C:\Users\User\PycharmProjects\exercitiu-git


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/28/2026   8:28 AM                depanare


(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git> cd depanare
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git add depanare
fatal: pathspec 'depanare' did not match any files
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git add depanare
fatal: pathspec 'depanare' did not match any files
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git add .
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -m "Pornire: inventar.py stricat"
[main 16ed0be] Pornire: inventar.py stricat
 2 files changed, 98 insertions(+)
 create mode 100644 depanare/inventar.py
 create mode 100644 depanare/tinte.json
```

Figura 2.1: **B1. Pregătire, în depozitul exercitiu-git**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 8
    def incarca(cale)
                    ^
SyntaxError: expected ':'
```

Figura 2.1: **Prima eroare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -am "SyntaxError: adaug : la definirea functiei incarca"
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it
[main 3a82642] SyntaxError: adaug : la definirea functiei incarca
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Figura 2.2: **Prima Rezolvare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 19
    return None
    ^^^^^^
IndentationError: expected an indented block after 'except' statement on line 18
```

Figura 2.3: **A doua Eroare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -am "IndentationError: indentez return None in port_valid"
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it
[main 458523b] IndentationError: indentez return None in port_valid
 1 file changed, 2 insertions(+), 2 deletions(-)
```

Figura 2.4: **A doua Rezolvare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
Traceback (most recent call last):
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 67, in <module>
    main()
    ~~~~^^
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 52, in main
    print("Tinte citite: " + len(tinta))
                               ^^^^^
NameError: name 'tinta' is not defined. Did you mean: 'tinte'?
```

Figura 2.5: **A treia Eroare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -am "NameError tinta: numele corect e tinte"
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it
[main 6f6ba3a] NameError tinta: numele corect e tinte
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Figura 2.6: **A treia Rezolvare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
Traceback (most recent call last):
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 67, in <module>
    main()
    ~~~~^^
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 52, in main
    print("Tinte citite: " + len(tinte))
          ~~~~~~~~~~~~~~~~~^~~~~~~~~~~~
TypeError: can only concatenate str (not "int") to str
```

Figura 2.7: **A patra Eroare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -am "TypeError: f-string pentru afisare tinte citite"
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it
[main 8e7d0a0] TypeError: f-string pentru afisare tinte citite
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Figura 2.8: **A patra Rezolvare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
Tinte citite: 6
sar peste 192.168.56.103 : port invalid abc
Traceback (most recent call last):
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 67, in <module>
    main()
    ~~~~^^
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 54, in main
    grupe = pe_servicii(tinte)
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 26, in pe_servicii
    serviciu = t["serviciu"]
               ~^^^^^^^^^^^^
KeyError: 'serviciu'
```

Figura 2.9: **A cincea Eroare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -am "KeyError serviciu: folosesc get cu valoare implicita necunoscut"
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it
[main aed8adc] KeyError serviciu: folosesc get cu valoare implicita necunoscut
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Figura 2.10: **A cincea Rezolvare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
Tinte citite: 6
sar peste 192.168.56.103 : port invalid abc
Traceback (most recent call last):
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 67, in <module>
    main()
    ~~~~^^
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 55, in main
    for serviciu, porturi in grupe.items():
                             ^^^^^^^^^^^
AttributeError: 'NoneType' object has no attribute 'items'
```

Figura 2.11: **A sasea Eroare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -am "AttributeError pe_servicii: adaug return grupe"
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it
[main 91fc67e] AttributeError pe_servicii: adaug return grupe
 1 file changed, 2 insertions(+)
```

Figura 2.12: **A sasea Rezolvare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
Tinte citite: 6
sar peste 192.168.56.103 : port invalid abc
ssh: [22]
http: [8080]
mysql: [3306]
necunoscut: [5432]
Traceback (most recent call last):
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 69, in <module>
    main()
    ~~~~^^
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 60, in main
    mari = peste_prag(tinte, a.prag)
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 42, in peste_prag
    if port > prag:
       ^^^^^^^^^^
TypeError: '>' not supported between instances of 'int' and 'str'
```

Figura 2.13: **A saptea Eroare**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -am "TypeError la prag: type=int in argparse"
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it
[main 350cfb3] TypeError la prag: type=int in argparse
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Figura 2.14: **A saptea Rezolvare**

```
grupe[serviciu] = []
grupe[serviciu].append(port)
```

Figura 2.15: **Eroare asunsa 1**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -am "Greseala tacuta: initializez lista doar daca serviciul nu exista"
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it
[main 2233693] Greseala tacuta: initializez lista doar daca serviciul nu exista
 1 file changed, 2 insertions(+), 1 deletion(-)
```

Figura 2.16: **Rezolvare a erorii ascuse 1**

```
if serviciu not in grupe:
    grupe[serviciu] = []
grupe[serviciu].append(port)
```

Figura 2.17: **Rezolvare in cod a erorii ascunse 1**

```
def peste_prag(tinte :{__iter__}, prag):  1 usage  Zlatin Nichita *
    """Lista gazdelor cu un port mai mare decat pragul."""
    rezultat = []
    for t in tinte:
        port = int(t["port"])
        if port > prag:
            rezultat.append(t["gazda"])
            return rezultat
```

Figura 2.18: **Eroare asunsa 2**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -am "Greseala tacuta: scot return din bucla for in peste_prag"
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it
[main c691117] Greseala tacuta: scot return din bucla for in peste_prag
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Figura 2.19: **Rezolvare a erorii ascunse 2**

```python
def peste_prag(tinte :{__iter__} , prag):  1 usage  Zlatin Nichita
    """Lista gazdelor cu un port mai mare decat pragul."""
    rezultat = []
    for t in tinte:
        port = int(t["port"])
        if port > prag:
            rezultat.append(t["gazda"])
    return rezultat
```

Figura 2.20: **Rezolvare in cod a erorii ascunse 2**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
Tinte citite: 6
sar peste 192.168.56.103 : port invalid abc
ssh: [22]
http: [80, 8080]
mysql: [3306]
necunoscut: [5432]
Traceback (most recent call last):
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 70, in <module>
    main()
    ~~~~^^
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 61, in main
    mari = peste_prag(tinte, a.prag)
  File "C:\Users\User\PycharmProjects\exercitiu-git\depanare\inventar.py", line 42, in peste_prag
    port = int(t["port"])
ValueError: invalid literal for int() with base 10: 'abc'
```

Figura 2.21: **Eroare aparuta**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -am "ValueError in peste_prag: folosesc port_valid pentru a trata porturile invalide"
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it
[main fa9a1f0] ValueError in peste_prag: folosesc port_valid pentru a trata porturile invalide
 1 file changed, 2 insertions(+), 2 deletions(-)
```

Figura 2.22: **Rezolvare a erorii aparute**

```python
def peste_prag(tinte :{__iter__}, prag):  1 usage  Zlatin Nichita
    """Lista gazdelor cu un port mai mare decat pragul."""
    rezultat = []
    for t in tinte:
        port = port_valid(t["port"])
        if port is not None and port > prag:
            rezultat.append(t["gazda"])
    return rezultat
```

Figura 2.23: **Rezolvare in cod a erorii aparute**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
Tinte citite: 6
sar peste 192.168.56.103 : port invalid abc
ssh: [22]
http: [80, 8080]
mysql: [3306]
necunoscut: [5432]
Peste 1000: ['192.168.56.102', '192.168.56.103', '192.168.56.104']
Raport salvat in raport_inventar.json
```

Figura 2.24: **Raportul rulat correct**

```
(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git add raport_inventar.json

(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git commit -m "Raportul uneltei reparate"
[main 3a2a197] Raportul uneltei reparate
 1 file changed, 22 insertions(+)
 create mode 100644 depanare/raport_inventar.json

(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare> git push
Enumerating objects: 50, done.
Counting objects: 100% (50/50), done.
Delta compression using up to 16 threads
Compressing objects: 100% (49/49), done.
Writing objects: 100% (49/49), 5.56 KiB | 813.00 KiB/s, done.
Total 49 (delta 22), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (22/22), completed with 1 local object.
To https://github.com/zlatinnichita2006-pixel/exercitiu-git.git
   53027d6..3a2a197  main -> main

(.venv) PS C:\Users\User\PycharmProjects\exercitiu-git\depanare>
```

Figura 2.25: **Salvarea modificarilor in git**

Concluzia: În urma efectuării acestei lucrări de laborator, s-a realizat cu succes depanarea și optimizarea scriptului `inventar.py`, fiind identificate și remediate toate cele 10 erori de sintaxă, de rulare și de logică. Prin utilizarea sistemului de control al versiunilor Git, fiecare reparație a fost salvată printr-un commit individual, transformând istoricul proiectului într-un jurnal structurat al procesului de depanare. În final, programul a rulat conform cerințelor, generând corect fișierul `raport_inventar.json`, iar istoricul complet al modificărilor a fost sincronizat pe platforma GitHub prin comanda `git push`.